Cambridge Mathematics Worksheet

Topically Structured Problem-Solving: Applied Percentages

Instructions: Answer all questions. Show all working clearly. Unless stated otherwise, give non-exact numerical answers correct to 3 significant figures or 1 decimal place for percentages.

1. A local bakery increased the price of a sourdough loaf from £3.20 to £3.68. Calculate the percentage increase in the price of the loaf.

2. During a seasonal promotion, a laptop originally priced at $850 is reduced by 15%.

(a) Calculate the discount amount in dollars.

(b) Find the final sale price of the laptop.

3. A jacket is sold in a store for $102 after a discount of 15% was applied to its original tag price. Calculate the original price of the jacket before the discount.

4. Marcus invests $2,400 in a savings account that pays a simple interest rate of 3.5% per annum. Calculate the total amount of money in his account after 4 years.

5. An electronics trader buys a television for £450 and sells it later for £558. Calculate his percentage profit.

6. The total cost of an online service including a 20% Value Added Tax (VAT) is £144. Find the cost of the service before VAT was added.

7. The population of a town was 45,000 in 2020.

(a) Between 2020 and 2022, the population increased by 8%. Calculate the population in 2022.

(b) Between 2022 and 2024, the population decreased by 5%. Calculate the population in 2024.

8. In a survey of 320 students:

(a) 45% walk to school. Calculate how many students walk to school.

(b) $\frac{3}{8}$ of the remaining students travel by bus. Calculate the percentage of the total survey group who travel by bus.

9. Sarah earns a monthly salary of $3,600. She spends 30% of her salary on rent, 25% on groceries, and saves $\frac{2}{5}$ of the remainder.

(a) Calculate the total amount Sarah spends on rent and groceries.

(b) How much money does Sarah save each month?

10. Hannah deposits $5,000 into a bank account offering compound interest at a rate of 4% per year. Calculate the total amount of money in her account at the end of 3 years.

11. Maurice claims: "If an item increases in price by 20% and then later decreases by 20%, the price returns to its original amount."

(a) By testing with an initial price of \$100, prove mathematically that Maurice is incorrect.

(b) Calculate the overall percentage change relative to the original price.

12. Following a successful marketing campaign, a company's annual revenue increased by 35% to \$1,485,000. Calculate the revenue of the company before the campaign.

13. A new car was bought for £22,000. It depreciates in value at a constant rate of 12% per year.

(a) Calculate the value of the car after 1 year.

(b) Calculate the value of the car after 3 years. Give your answer to the nearest pound.

14. A collector bought a rare painting for \$12,500 and sold it five years later for \$10,750. Calculate her percentage loss.

15. A sports field measures 120 metres by 80 metres. A groundskeeper fertilises 65% of the total area of the field. Calculate the area of the field, in square metres, that has not been fertilised.

16. Store A and Store B sell the same model of bicycle, originally priced at \$400.

Store A offers a straight discount of 25%.
Store B offers a discount of \$60 followed by an additional 10% off the reduced price.

Determine which store offers the lower final price and calculate the price difference.

17. An engineer estimated the length of a bridge to be 480 metres. Upon exact measurement, the actual length of the bridge was found to be 500 metres. Calculate the percentage error of the engineer's estimate.

18. A metal alloy weighing 850 grams is made of 64% copper, 28% zinc, and the rest is nickel.

(a) Calculate the mass of nickel in the alloy, in grams.

(b) Express the ratio of copper to zinc in its simplest integer form.

19. Daniel earns a gross annual income of £42,000.

(a) The first £12,500 of his income is tax-free. Calculate his taxable income.

(b) He pays income tax at a rate of 20% on all taxable income. Calculate the total income tax he pays in a year.

(c) Express his tax payment as a percentage of his total gross income.

20. A factory produced 15,000 units of a product in January.

(a) In February, production increased by 12%. How many units were produced in February?

(b) In March, 4.5% of the total February production was found to be defective. Calculate the number of non-defective units produced in February.

(c) If each non-defective unit is sold for \$8.50, calculate the total revenue generated from the non-defective units produced in February.